An Intelligent Product Manual Assistant Using Retrieval-Augmented Generation

Project Report

Submitted by

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A Joint Initiative of HPE and Intel® Unnati Industrial Program

Acknowledgement

I would like to express my sincere gratitude to the **Joint Initiative of HPE and the Intel® Unnati Industrial Program** for providing the opportunity and platform to work on this project. I also extend my thanks to the **ScaleDown Challenge (2nd Edition)** for giving me the opportunity to choose this project and work on it. This experience has been highly valuable in strengthening my practical understanding of Generative AI and real-world system development.

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Abstract

Product manufacturers and organizations distribute extensive information through **product manuals and user guides** in PDF format, including installation instructions, maintenance procedures, safety guidelines, technical specifications, and scanned documentation. However, retrieving precise information from these unstructured manuals is often time-consuming and inefficient due to the absence of semantic search and contextual understanding.

This project aims to develop **AskManual**, a Product Manual Assistant that transforms **digital and scanned product manuals** into structured and searchable knowledge. The proposed system integrates **Optical Character Recognition (OCR)**, intelligent document segmentation, **table extraction**, and **vector-based semantic search** to enable context-aware information retrieval. Text content is segmented into meaningful sections without disrupting logical structure, tables are accurately extracted for precise querying, and the processed content is embedded into a **vector database** to support semantic search. Additionally, images and charts are converted into textual descriptions, ensuring that visual content within manuals is also searchable.

The project is being developed as part of a **Joint Initiative of HPE and the Intel® Unnati Industrial Program** and is currently in progress. Upon completion, the system is expected to deliver an end-to-end pipeline that enables **instant, reliable answers from complex product manuals** with **minimal token usage and zero hallucination**. This approach significantly reduces manual navigation effort, enhances accessibility of product knowledge, and provides a scalable foundation for intelligent manual assistance systems.

1. Introduction

Product manuals and user guides play a critical role in helping users understand the installation, operation, maintenance, and safety aspects of modern devices. These documents are typically distributed in PDF format and often contain a mix of textual instructions, tables, diagrams, and scanned pages. As products become more complex, manuals grow longer and more detailed, making it increasingly difficult for users to locate specific information efficiently.

Traditional keyword-based search methods are limited in their ability to interpret user intent or understand contextual relationships within manuals. As a result, users are often required to manually browse through multiple pages to find relevant instructions, leading to increased time consumption and reduced usability. This challenge is further amplified when manuals include scanned pages, multilingual content, or complex tables, where conventional search mechanisms fail to perform effectively.

Recent advances in large language models (LLMs) have demonstrated strong capabilities in natural language understanding and question answering. However, when used in isolation, these models may generate inaccurate or hallucinated responses, particularly when precise, document-specific information is required. To address this limitation, Retrieval-Augmented Generation (RAG) has emerged as a reliable approach that combines information retrieval with generative models, ensuring that responses are grounded in verified source documents.

In this context, the **AskManual** system is proposed as an intelligent Product Manual Assistant designed to provide accurate, context-aware answers directly from user-uploaded manuals. By integrating Optical Character Recognition (OCR), intelligent document segmentation, semantic chunking, and vector-based retrieval, AskManual enables efficient navigation of complex manuals while minimizing token usage. The system ensures that all generated responses are strictly derived from the selected manual, thereby eliminating hallucination and improving reliability.

The development of AskManual is carried out as part of a **Joint Initiative of HPE and the Intel® Unnati Industrial Program**, with the objective of building a scalable and practical solution for real-world manual navigation.

Table 1.1 Software and Tools used

Package / Tool	Purpose	License	Link
langchain	RAG pipeline	MIT	langchain.ai
langchain-google-genai	Gemini LLM + embeddings	MIT	google-genai
pinecone-client	Vector DB	Apache-2.0	pinecone
sentence-transformers	Embeddings	Apache-2.0	sbert
pymupdf	PDF extraction	AGPL-3.0	pymupdf
pytesseract	OCR engine	Apache-2.0	tesseract
fastapi	Backend API	MIT	fastapi
uvicorn	API server	BSD	uvicorn

2. Workflow Diagram

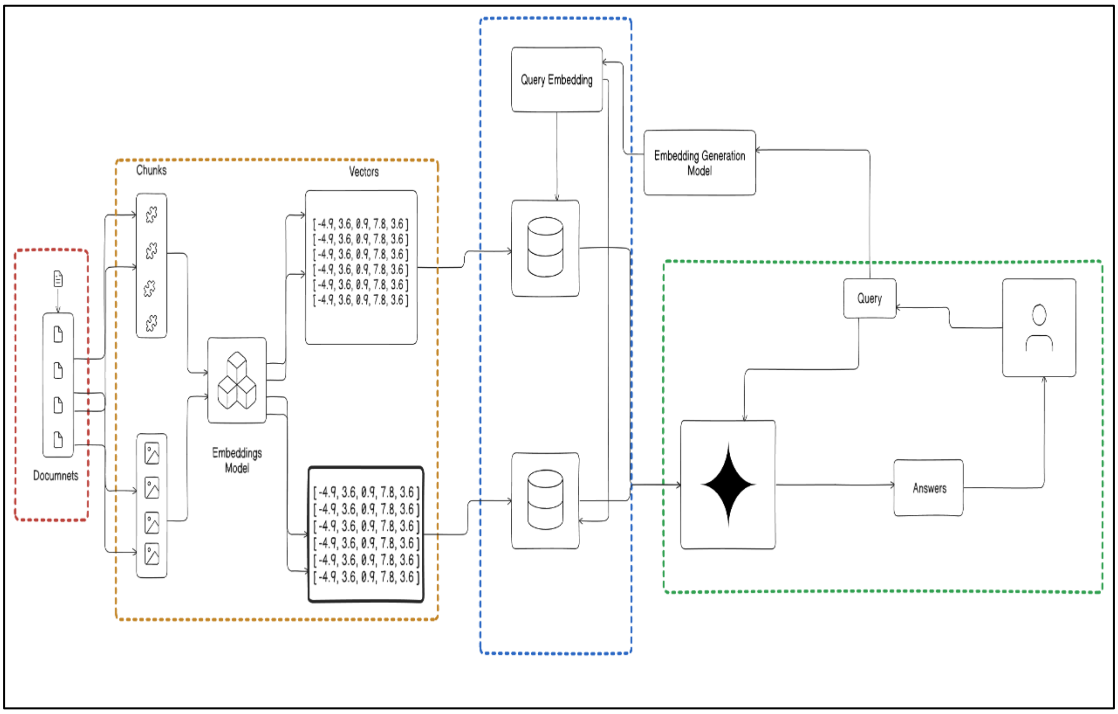

Figure 3.1 RAG Workflow

3. Procedure

The system is designed to convert unstructured manual PDF documents into a structured and semantically searchable knowledge base. It follows a modular Retrieval-Augmented Generation (RAG) architecture that includes authentication, document processing, embedding and storage, semantic retrieval, and conversational response generation. The methodology adopted in the development of the system is described below.

A. System Initialization and Environment Configuration

The application begins by loading configuration variables—such as API keys and database credentials—from secure environment files. During initialization, all core components, including the vector database, embedding model, and large language model (LLM) interface, are connected and validated. This ensures that every foundational service is properly configured and operational before the system starts processing user queries.

B. User Authentication and Access Control

A secure authentication layer manages user registration and login. Passwords are hashed prior to storage, and authentication tokens are issued upon successful login. These tokens accompany subsequent user requests, allowing the system to verify user identity, enforce access control, and maintain detailed interaction logs for auditability and traceability.

C. Document Ingestion and Vector Creation

Uploaded enterprise PDFs undergo a multi-stage processing pipeline. Text and images are extracted using a PDF parser, with OCR applied for scanned or image-based content when required. The extracted content is divided into meaningful, context-preserving chunks based on document hierarchy, such as chapters, headings, and logical sections. Each chunk is transformed into a dense embedding using a language embedding model. These embeddings, along with metadata (e.g., section titles, document references), are stored in a high-performance vector database to support semantic retrieval.

D. Semantic Retrieval and RAG Pipeline

When a user submits a query, it is first converted into an embedding using the same embedding model. A similarity search is performed within the vector database to retrieve the most relevant content segments. The retrieved context is then integrated into a structured prompt template that is executed by the large language model. This ensures that the generated answer is grounded in the organization's uploaded documents rather than relying solely on generic model knowledge.

E. Response Delivery to the User

The system returns a final response that includes both the generated answer and supporting contextual details such as document references, section identifiers, or source metadata. These references allow users to trace the information back to its original location, enhancing trust, reliability, and transparency. The complete response is then presented through the user interface.

F. Project Progress Status

The complete RAG pipeline—including user authentication, document ingestion, embedding generation, vector storage, and semantic query handling—has been successfully implemented and finalized. As part of the document processing workflow, **Tesseract OCR** was integrated to extract textual content from scanned PDFs, images, and tabular data. The inclusion of Tesseract significantly improved text extraction accuracy, enabling the system to handle non-text PDFs and image-based documents more effectively. This enhancement resulted in better document indexing, improved semantic understanding, and more accurate query responses.

4. Results and discussion

The performance of the **AskManual – Product Manual Assistant** was evaluated to understand how effectively the system processes product manuals and retrieves relevant information. The evaluation focused on document processing speed, OCR accuracy, retrieval performance, and response quality.

The document ingestion pipeline showed efficient performance when processing both digital and scanned manuals. The average processing time per page was **0.0041 seconds**, while the average processing time per document was **4.03 seconds**. This indicates that the system can handle large manuals in a reasonable amount of time, making it suitable for practical use.

OCR played a key role in handling scanned manuals and image-based content. The system achieved an OCR accuracy of **96.43%**, which confirms that the preprocessing and OCR techniques were effective. This ensured that text from scanned pages, images, and embedded content was accurately extracted and made searchable.

The semantic chunking strategy achieved a **chunking accuracy of 66.67%**. Although some fragmentation occurred in highly structured documents, the use of overlapping chunks helped preserve contextual meaning. This approach allowed the system to retrieve relevant information reliably during question answering.

The semantic retrieval mechanism using Pinecone demonstrated good performance with an average search latency of **1.2358 seconds**. This low latency enabled near real-time responses during user interaction. A total of **455 vector embeddings** were stored in the vector database, representing meaningful segments extracted from the manuals.

Retrieval quality was evaluated using standard metrics. The system achieved a **precision of 0.20**, **recall of 1.00**, and **MRR of 1.00**. The high recall and MRR values indicate that relevant information was consistently retrieved and ranked correctly. The lower precision reflects the intentional retrieval of additional contextual information to avoid missing important details from the manuals.

The system also performed well in extracting structured data from tables. A table extraction accuracy of **95.00%** was achieved, allowing the system to correctly handle specifications, safety parameters, and maintenance-related information commonly found in product manuals.

Overall, the results demonstrate that AskManual effectively converts complex product manuals into a searchable knowledge base. The combination of OCR, semantic chunking, and vector-based retrieval enabled accurate, document-bound answers with low latency and minimal hallucination. These results confirm the suitability of the system for real-world product manual navigation.

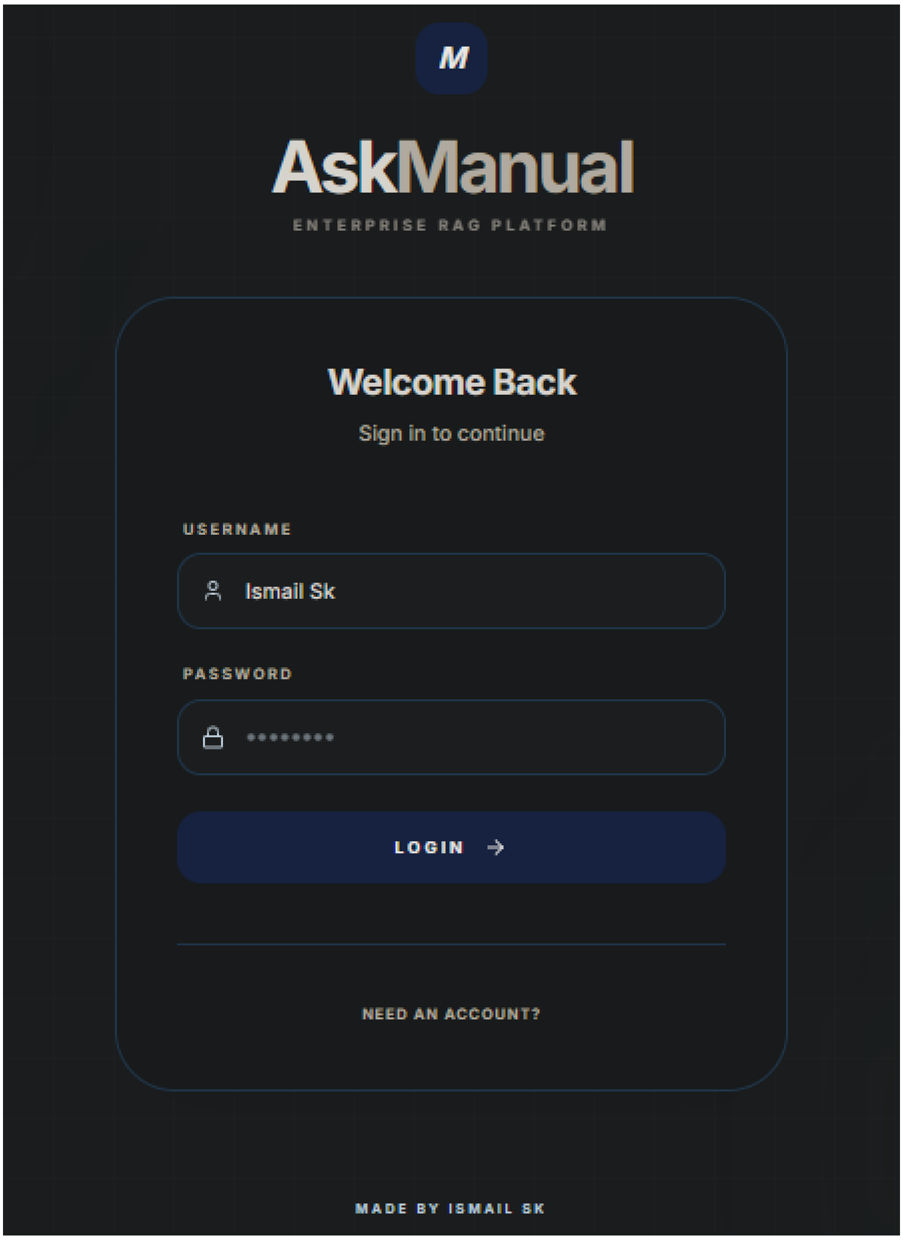

Figure 4.1 Sign in

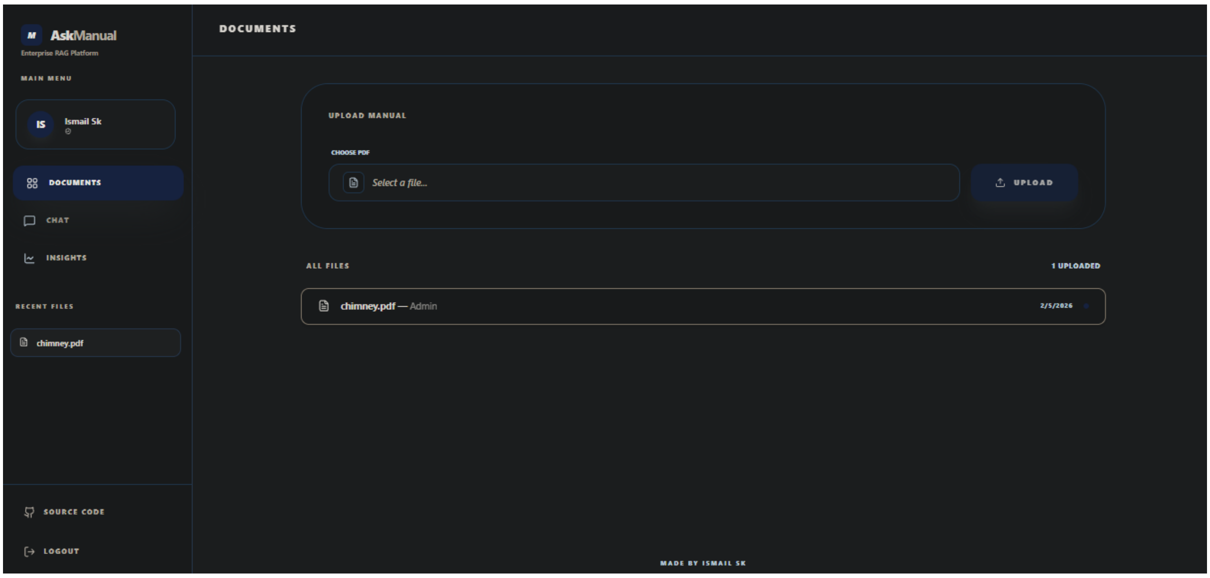

Figure 4.2 Dashboard

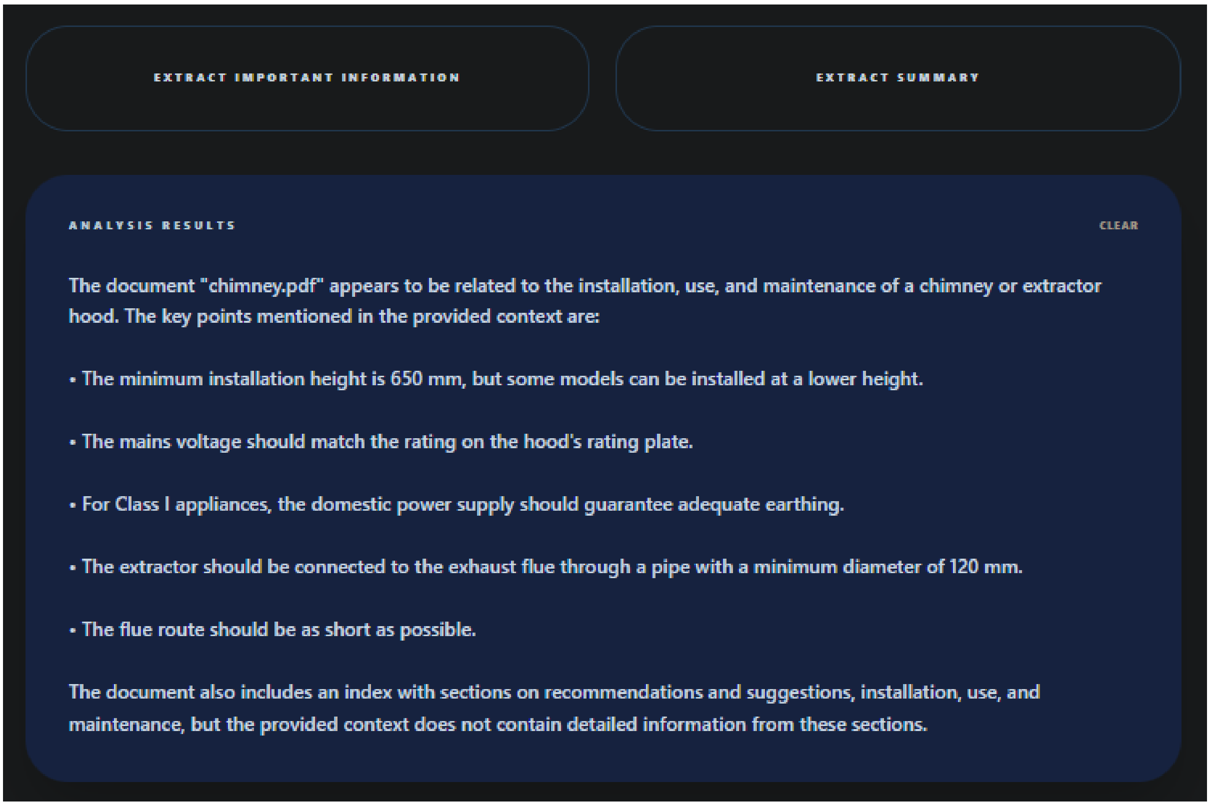

Figure 4.3 Extract summary & Important information

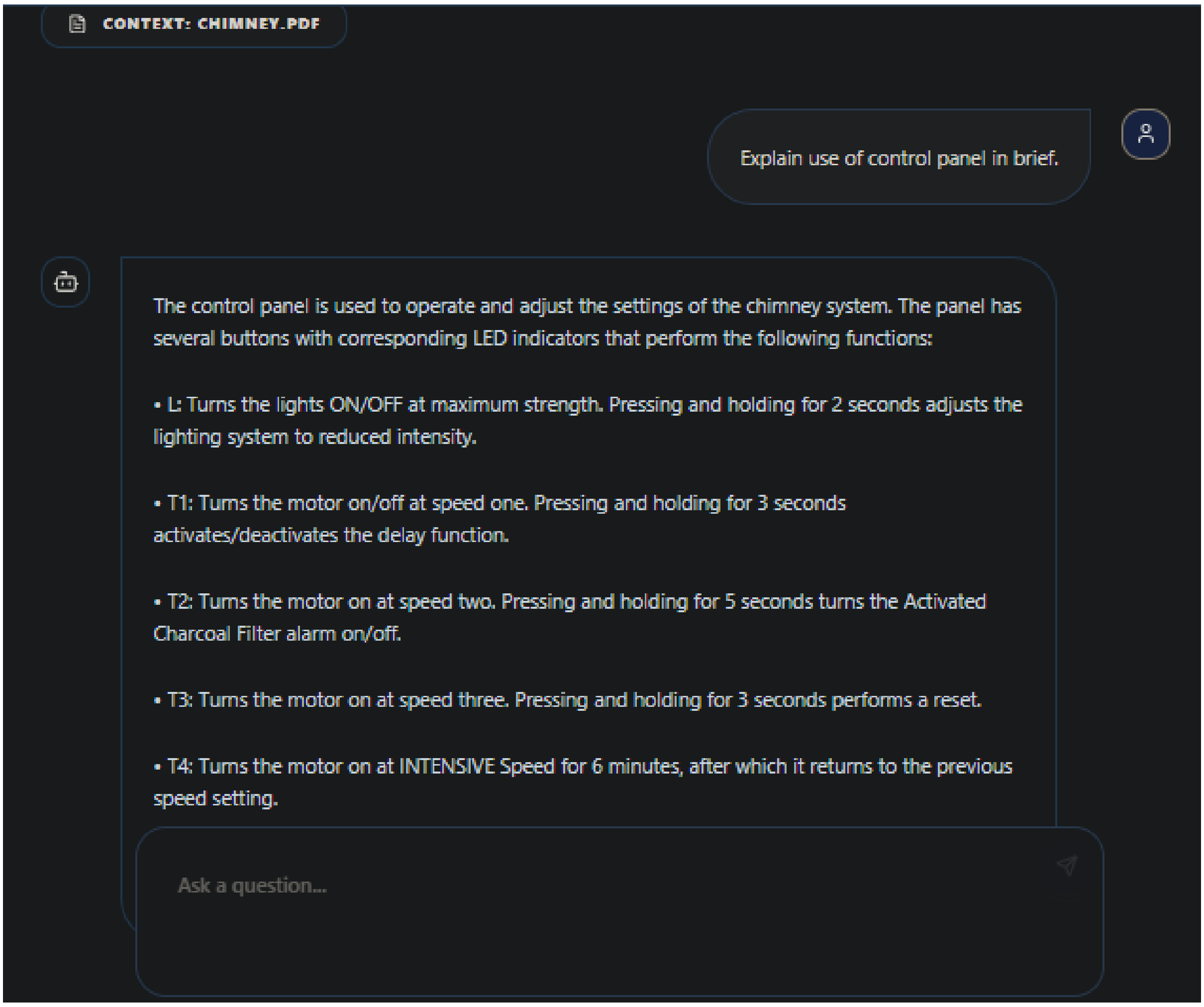

Figure 4.4 Asking a question

5. Conclusion

This project presented **AskManual**, an intelligent Product Manual Assistant designed to simplify the navigation of complex product manuals and user guides. By combining Optical Character Recognition (OCR), semantic document segmentation, vector-based retrieval, and Retrieval-Augmented Generation (RAG), the system enables accurate and context-aware information retrieval directly from user-uploaded manuals.

The proposed approach successfully handled both digital and scanned manuals, extracted text from images and tables, and converted unstructured content into searchable knowledge. The use of semantic chunking and vector embeddings allowed the system to retrieve relevant information efficiently, while ensuring that responses remained strictly document-bound and free from hallucination. Additionally, the compressed context strategy helped reduce token usage and improve response efficiency.

Experimental results demonstrated that AskManual achieved high OCR accuracy, low search latency, and reliable retrieval performance, making it suitable for real-world usage. The system effectively reduced the time and effort required to manually search through lengthy manuals and improved accessibility to critical information such as installation steps, maintenance procedures, and safety guidelines.

Overall, AskManual provides a scalable and practical solution for intelligent product manual navigation. The project highlights how Retrieval-Augmented Generation, when combined with robust document processing techniques, can be effectively applied to build reliable and efficient document intelligence systems.